Two-Phase Commit (2PC) and Consensus Protocols in Distributed Databases

Distributed transactions spanning multiple autonomous database partitions require atomic commitment protocols to ensure that either all participating nodes commit the transaction or all abort it (Atomicity). The classic Two-Phase Commit (2PC) protocol consists of a coordinator and multiple cohort participants.

Phase 1: Prepare Phase. The coordinator sends a Prepare message to all cohorts. Each cohort checks local constraints, locks necessary rows, flushes undo/redo log records to persistent storage, and votes YES (prepared) or NO (abort). Phase 2: Commit Phase. If every cohort voted YES, the coordinator logs a COMMIT record to disk and broadcasts Commit to cohorts. If any cohort voted NO or timed out, the coordinator broadcasts Abort. The primary limitation of standard 2PC is that it is a blocking protocol: if the coordinator crashes after cohorts enter the PREPARED state, cohorts must wait indefinitely for recovery.